

Sina Sajadmanesh

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Education

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering	May 2019 – Aug 2023
Sharif University of Technology, MSc in Information Technology Engineering	Sept 2014 – Sept 2016
University of Isfahan, BSc in Computer Software Engineering	Sept 2009 – Feb 2014

Experience

AI Engineer, Sony AI – Zurich, Switzerland Oct 2023 – present

- Working on privacy-preserving vision foundation models
- Implemented easy-to-use APIs for vision foundation model inference and deployment
- Developed a comprehensive multi-task learning framework based on OpenMMLab libraries and PyTorch, supporting various vision tasks, new model architectures, and mixed-precision and distributed training
- Implemented an end-to-end open-world classification pipeline using MMClassification, HuggingFace transformers, and CLIP
- Improved model training and inference speed by optimizing the data pipeline, using mixed-precision training, and leveraging distributed training on multiple GPUs

Research Assistant, Idiap Research Institute – Martigny, Switzerland May 2019 – Aug 2023

- Worked as a doctoral researcher on differentially private machine learning with graph neural networks
- **3 published papers** in top-tier conferences (CCS, USENIX Security, and NDSS) with **185+ citations**
- **7 invited talks** at top universities and research institutions, including Imperial College, UIC, and Twitter
- **6 open-source projects** with **110+ stars** on GitHub
- **1 short course** taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy
- **Finalist** in CSAW Applied Research Competition for the best paper award in computer security in Europe

Visiting Collaborator, The Alan Turing Institute – London, UK Mar 2023 – Mar 2023

- Co-organized a workshop on "Privacy and Fairness in AI for Health" with 60+ attendees

Research Intern, Brave Software – Remote Mar 2022 – May 2022

- Developed a novel privacy-preserving **federated learning** framework for **neural bandit models** under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite

Research Assistant, Sharif University of Technology – Tehran, Iran Nov 2014 – May 2019

- Worked on various research projects including privacy-preserving machine learning, web data science, and social and information network analysis.
- **4 published papers** in top-tier venues (WWW, TKDD, IoTJ, ASONAM) with **430+ citations**
- **5+ press releases** in top-tier media outlets, including MIT Technology Review, France 24, and The Independent
- **2 open-source projects** with **20+ stars** on GitHub
- **1 semester course** taught on "Fundamentals of Programming with Python" at Sharif University of Technology

Skills and Expertise

Programming Languages: Python, C++, Java, SQL, Shell, \LaTeX

Machine Learning and AI: PyTorch, TensorFlow, HuggingFace, OpenMMLab, PyTorch-Lightning

MLOps and DevOps: Weights & Biases, Docker, GitHub, Dask, Neptune, Linux

Privacy-Enhancing Technologies: Flower, Opacus, Auto-DP

Community and Professional Service

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), Graph Neural Networks User Group Meetup (2021), Twitter Machine Learning Seminar (2021)

Organizing Committee: Privacy and Fairness in AI for Health (2023)

Program Committee: PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2024), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

Publications

- Argus: A Compact and Versatile Foundation Model for Vision** June 2025
Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees** Mar 2024
Sina Sajadmanesh, Daniel Gatica-Perez
ACM International Conference on Web Search and Data Mining (WSDM)
- Privacy-Preserving Machine Learning on Graphs** Aug 2023
Sina Sajadmanesh
Doctoral Thesis - Swiss Federal Institute of Technology (EPFL)
- GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation** Aug 2023
Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, et al.
USENIX Security Symposium (USENIX Security)
- Locally Private Graph Neural Networks** Nov 2021
Sina Sajadmanesh, Daniel Gatica-Perez
ACM Conference on Computer and Communications Security (CCS)
- When Differential Privacy Meets Graph Neural Networks** June 2020
Sina Sajadmanesh, Daniel Gatica-Perez
Technical Report, ArXiv e-prints
- A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics** May 2020
Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al.
IEEE Internet of Things Journal (IoTJ)
- Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks** Aug 2019
Sina Sajadmanesh, Sogol Bazargani, Jiawei Zhang, Hamid R. Rabiee
ACM Transactions on Knowledge Discovery from Data (TKDD)
- NPGLM: A Non-Parametric Method for Temporal Link Prediction** June 2017
Sina Sajadmanesh, Jiawei Zhang, Hamid R. Rabiee
Technical Report, ArXiv e-prints
- Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web** Apr 2017
Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al.
International World Wide Web Conference Companion (WWW)
- Predicting Anchor Links between Heterogeneous Social Networks** Aug 2016
Sina Sajadmanesh, Hamid R. Rabiee, Ali Khodadadi
International Conference on Advances in Social Networks Analysis and Mining (ASONAM)